Week 4: Packet Tracer WLAN Configuration

RENE OLUOCH

CS-CNS09-25030

Contents

Introduction

In today's increasingly wireless world, understanding how to design and secure wireless LANs is fundamental to building robust network infrastructures. This lab focused on configuring WLANs in both home and enterprise environments using Cisco Packet Tracer. It provided hands-on experience in deploying wireless networks secured with WPA2-Personal and WPA2-Enterprise, implementing DHCP, RADIUS authentication, and SNMP services, and managing connectivity through both a home wireless router and a Wireless LAN Controller (WLC). By simulating real-world scenarios, this exercise not only reinforced critical networking concepts but also emphasized the importance of secure wireless configurations in modern network deployments.

Configuring DHCP Settings on the Home Wireless Router

To begin, I accessed the Home Wireless Router's GUI and made essential configuration changes to meet the network requirements. I changed the default LAN IP address to 192.168.6.1/27, aligning it with the provided subnet. Next, I modified the DHCP server settings to limit IP address allocation to a maximum of 20 hosts. I ensured the DHCP pool started from .3, so the usable range began at 192.168.6.3. For the WAN or Internet interface, I enabled DHCP, allowing the router to receive its IP dynamically. Upon applying the changes, the interface was successfully assigned an IP address from the upstream DHCP server (you can confirm the specific address if needed). Finally, I set the DNS server statically to 10.100.100.252, as specified in the Addressing Table. With all settings saved, the home network was now correctly configured to provide dynamic addressing and DNS resolution to internal clients.

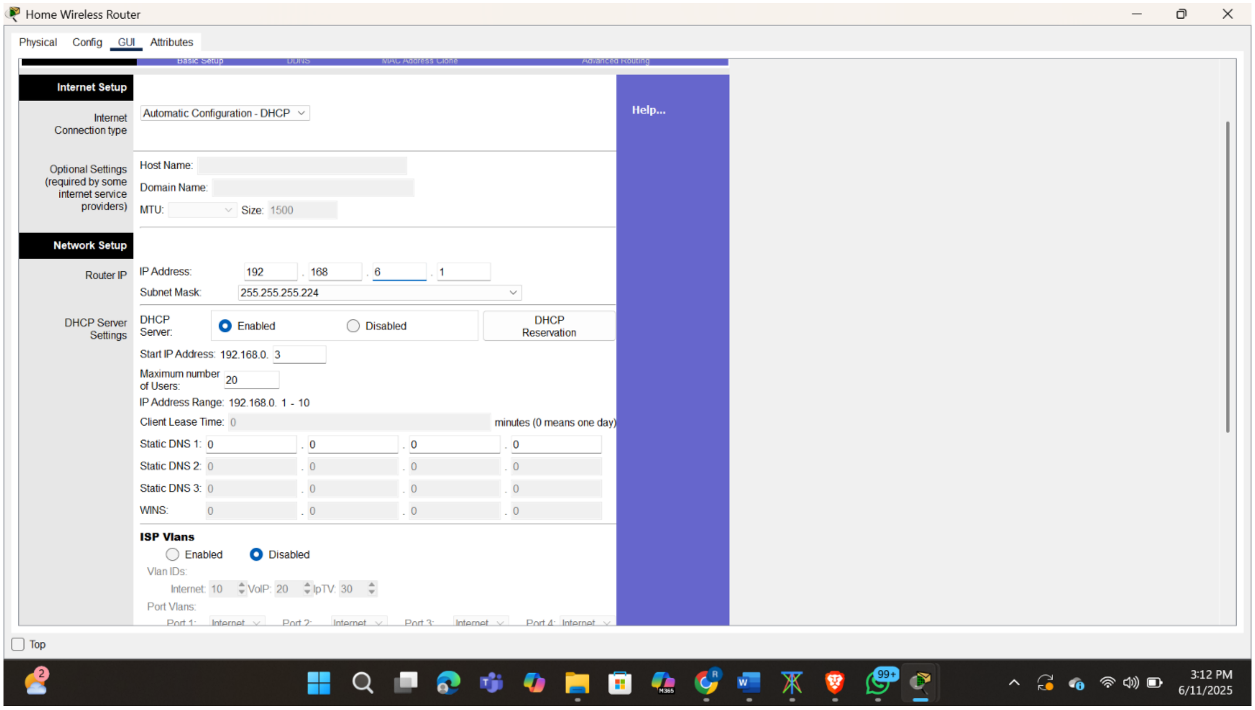

Fig 1.0 Changing home router settings

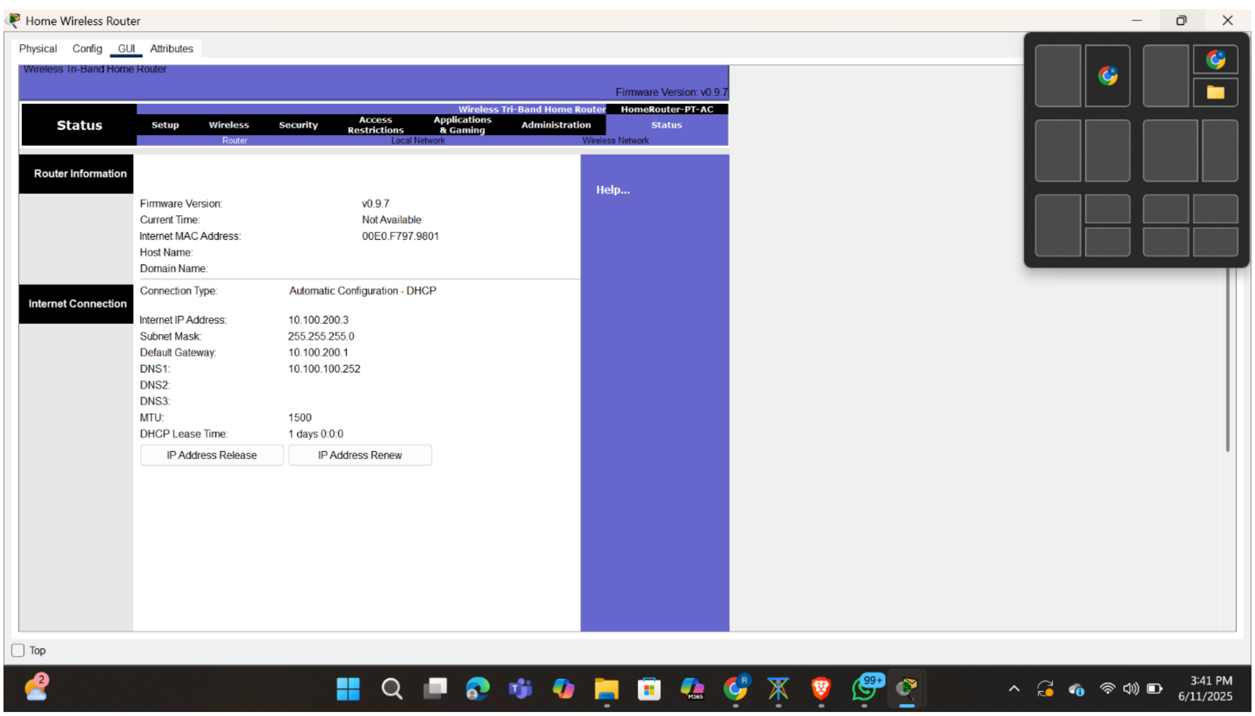

Fig 1.1 Verify the address home router received

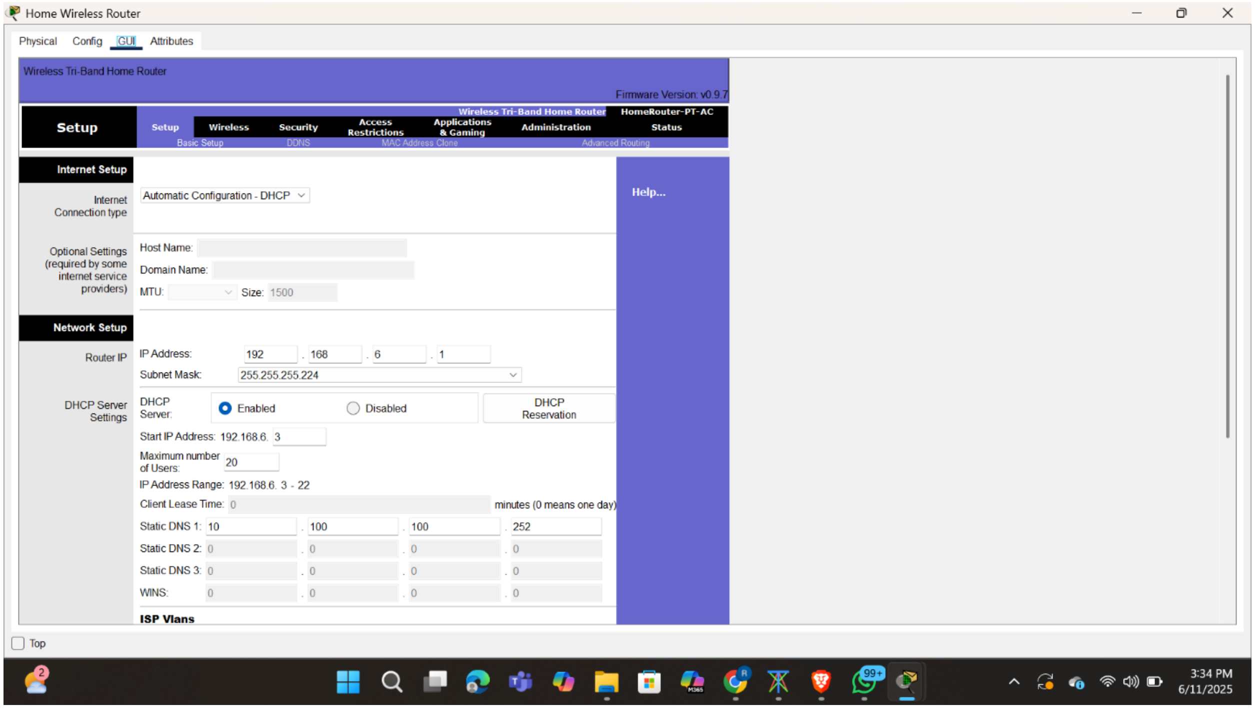

Fig 1.3 Configure the static DNS server to the address in the Addressing Table.

Configuring the Wireless LAN

After setting up DHCP on the Home Wireless Router, I proceeded to configure the wireless LAN. I selected the 2.4GHz band to provide wider compatibility for household devices. I assigned the SSID "HomeSSID" as per the WLAN Information Table, ensuring a recognizable and user-friendly network name. To minimize channel interference in the local area, I selected Channel 6, a commonly stable option for most environments. Crucially, I made sure that SSID broadcasting was enabled, allowing all Wi-Fi-capable hosts in the vicinity to detect and connect to the wireless network. These settings established a basic yet robust wireless environment ready for secure client connections.

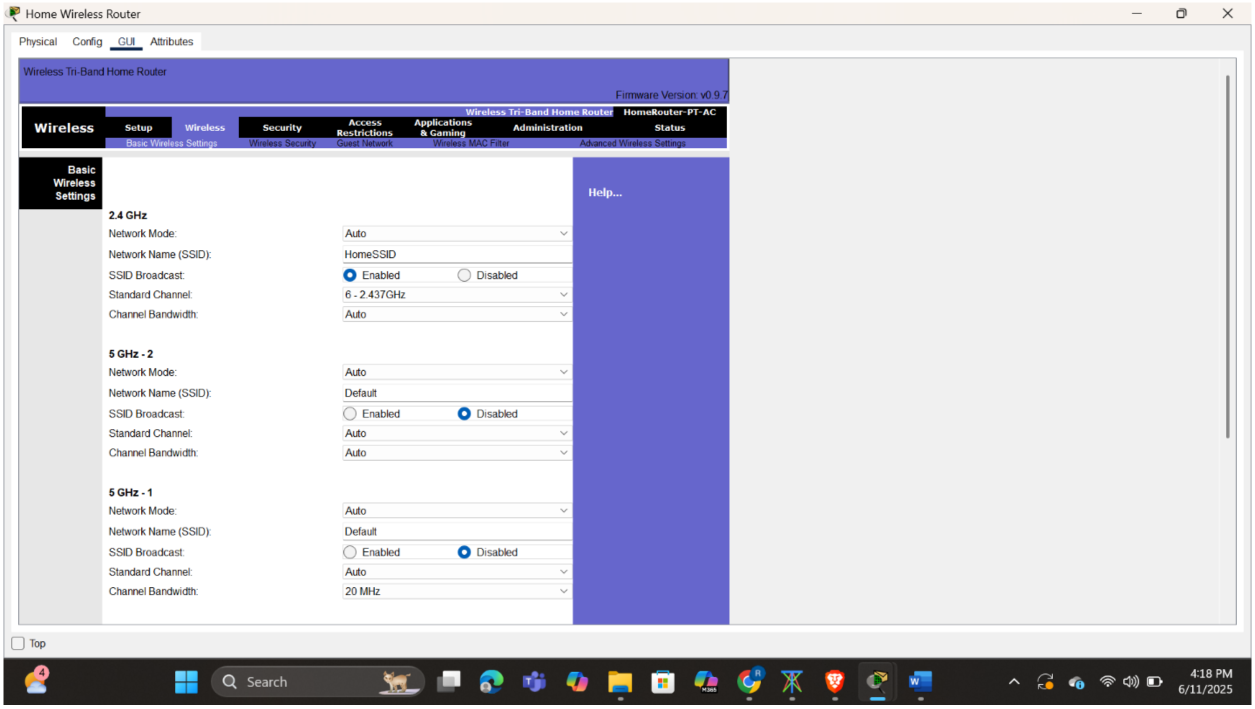

Fig 1.4 Setting up home network

Configuring security

To ensure the wireless network was secure, I implemented WPA2-Personal security on the 2.4GHz wireless LAN. This is a widely recommended encryption standard for home networks, balancing strong security with broad device compatibility. I used the passphrase "Cisco123", as specified in the WLAN Information table. This password will be required by any device attempting to join the wireless network, thereby helping prevent unauthorized access.

Next, I addressed administrative security by changing the router's default login credentials. I updated the router’s web interface password to "Cisco123", enhancing protection against unauthorized configuration changes. This step is often overlooked in home environments, yet it’s essential in preventing intrusions through the router management console.

At this point, the home wireless network is both operational and reasonably secure with encrypted access and protected administrative controls.

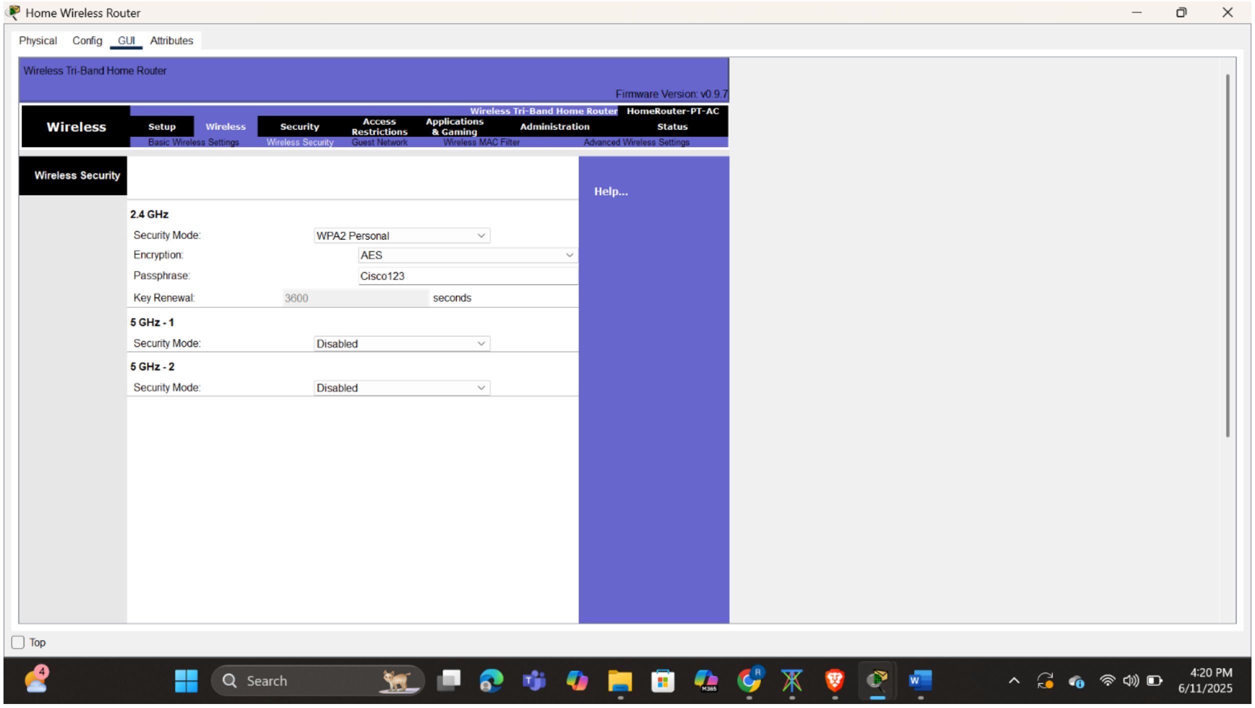

Fig 1.5 Home network security configuration

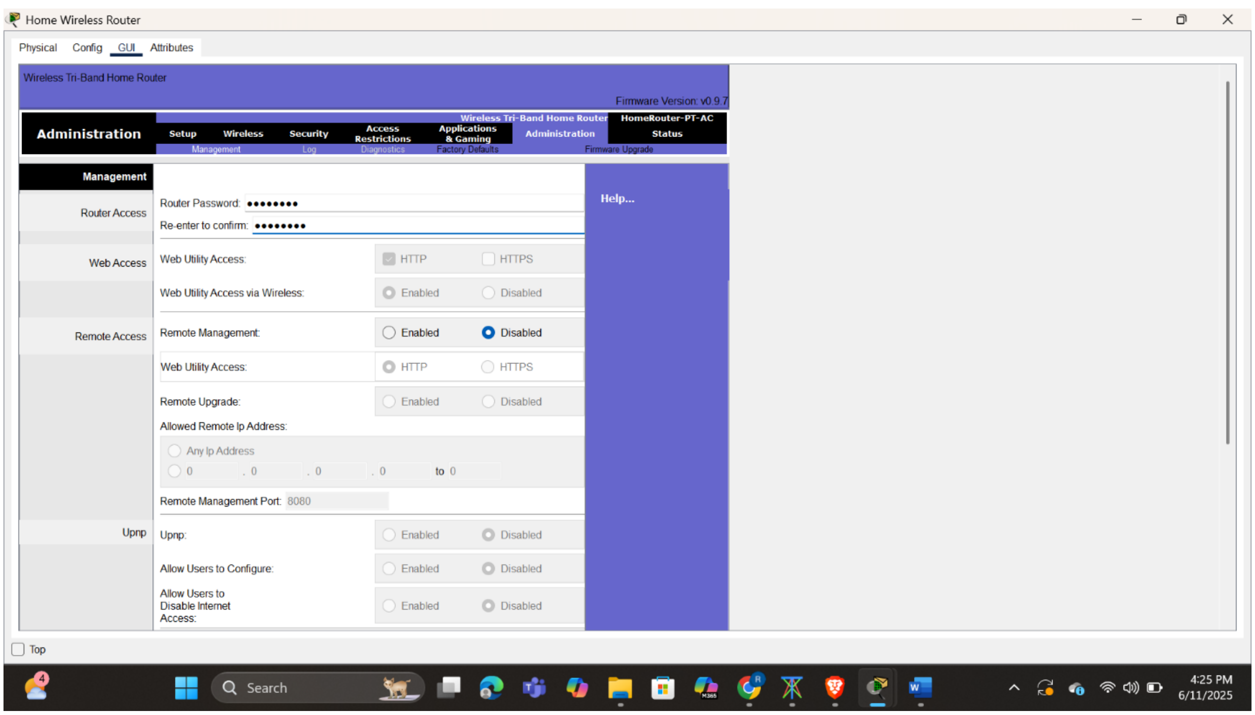

Fig 1.6 Changing routers default password

Connecting Clients to the Home Wireless Network

With the wireless network now properly configured and secured using WPA2-Personal, I began connecting the client devices to the HomeSSID network by first opening the PC Wireless application on the Laptop, scanning for available networks, selecting HomeSSID, and entering the passphrase “Cisco123” to authenticate; the Laptop successfully connected, obtained an IP address via DHCP, and joined the network. I then moved on to the Tablet PC and Smartphone, where I opened the Config tab, navigated to the Wireless0 interface, selected the same SSID, entered the same passphrase, and confirmed that both devices were authenticated and assigned IP addresses starting from 192.168.6.3 as expected based on the DHCP configuration. To verify connectivity across the network, I used the ping tool from each device to ensure they could communicate with one another on the WLAN, then confirmed external reachability by pinging the Web Server at 203.0.113.78, with all responses successfully returned. Finally, I tested access to the web server through its URL using a web browser, which confirmed both DNS resolution and functional internet access. At this point, all home wireless clients were securely connected, fully operational, and communicating as intended, thereby completing the configuration and validation of the home WLAN setup.

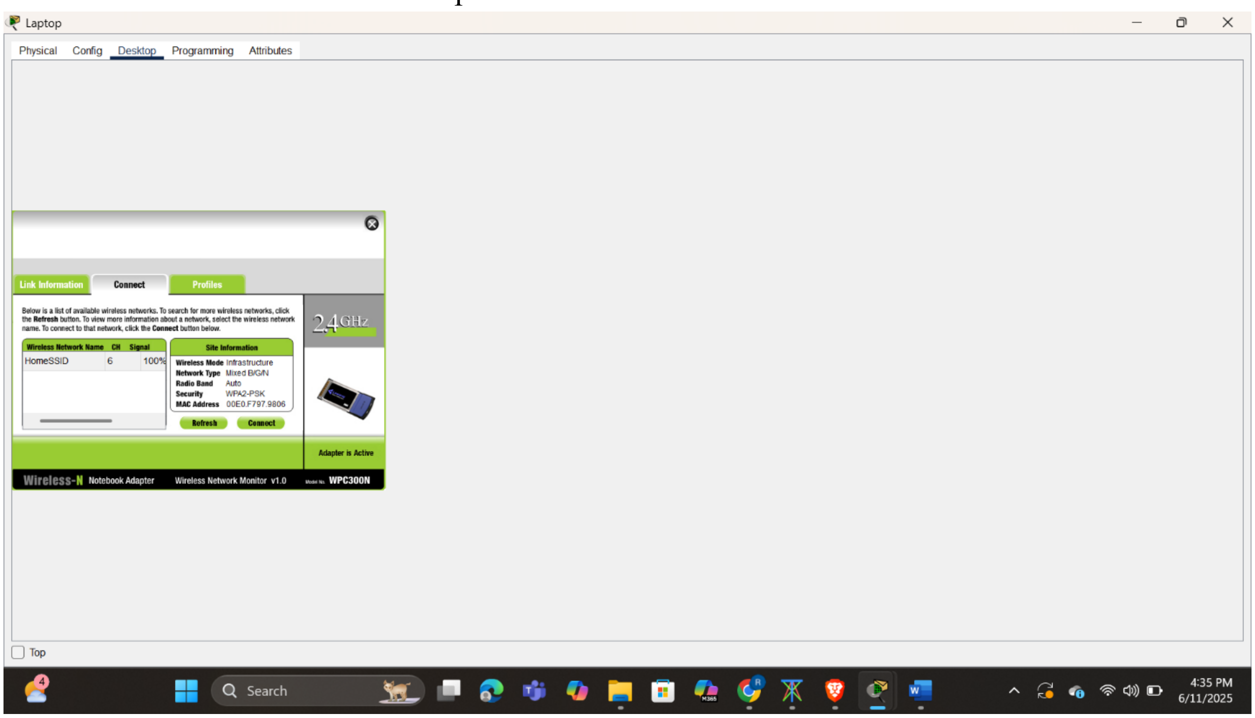

Fig 1.7 Connecting pc to wireless network

Opening the Config tab on the Tablet PC and Smartphone and configure the wireless interfaces to connect to the wireless network.

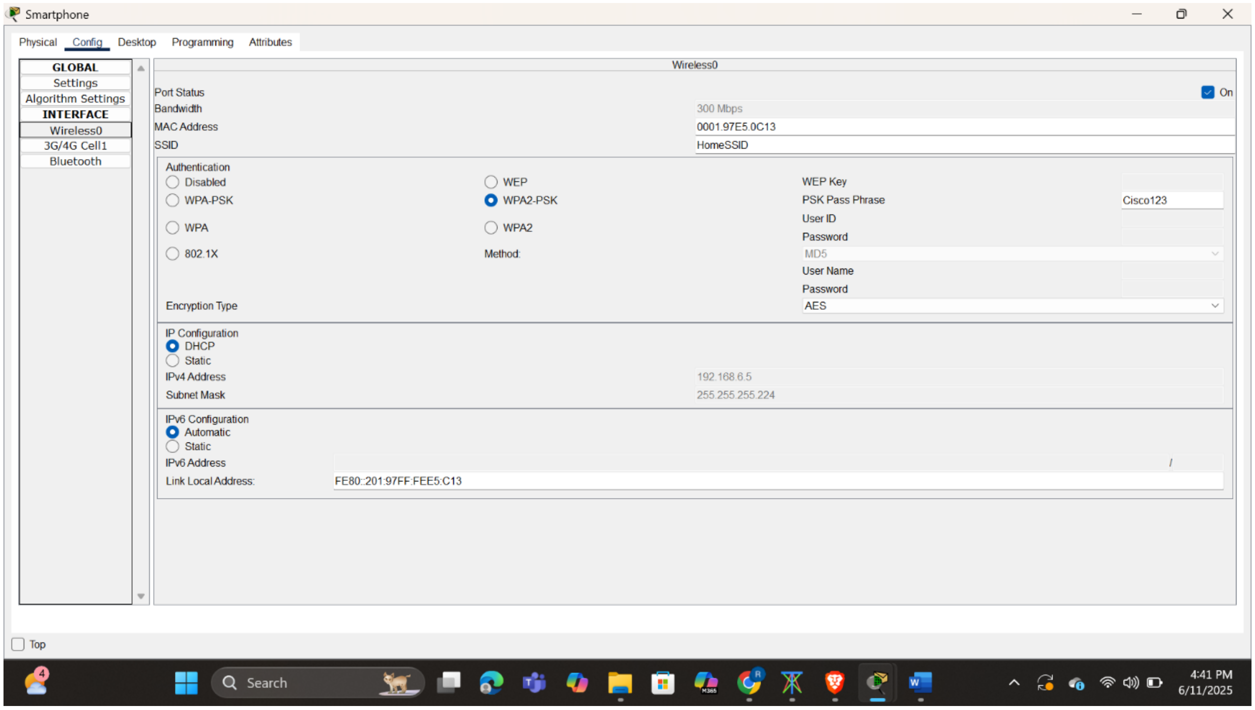

Fig 1.8 Connecting smartphone to wireless network

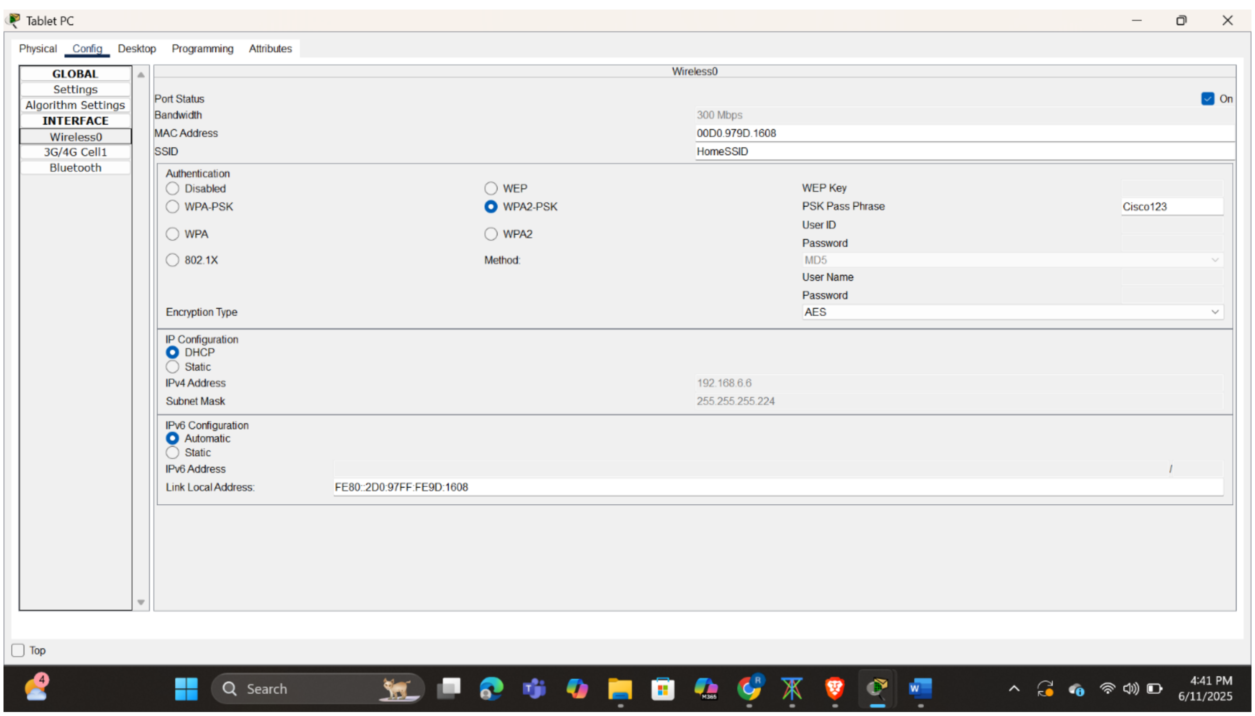

Fig 1.9 Connecting tablet pc to wireless connection

To verify connectivity, I was able to ping each device and the web server. They were also able to reach the web server URL.

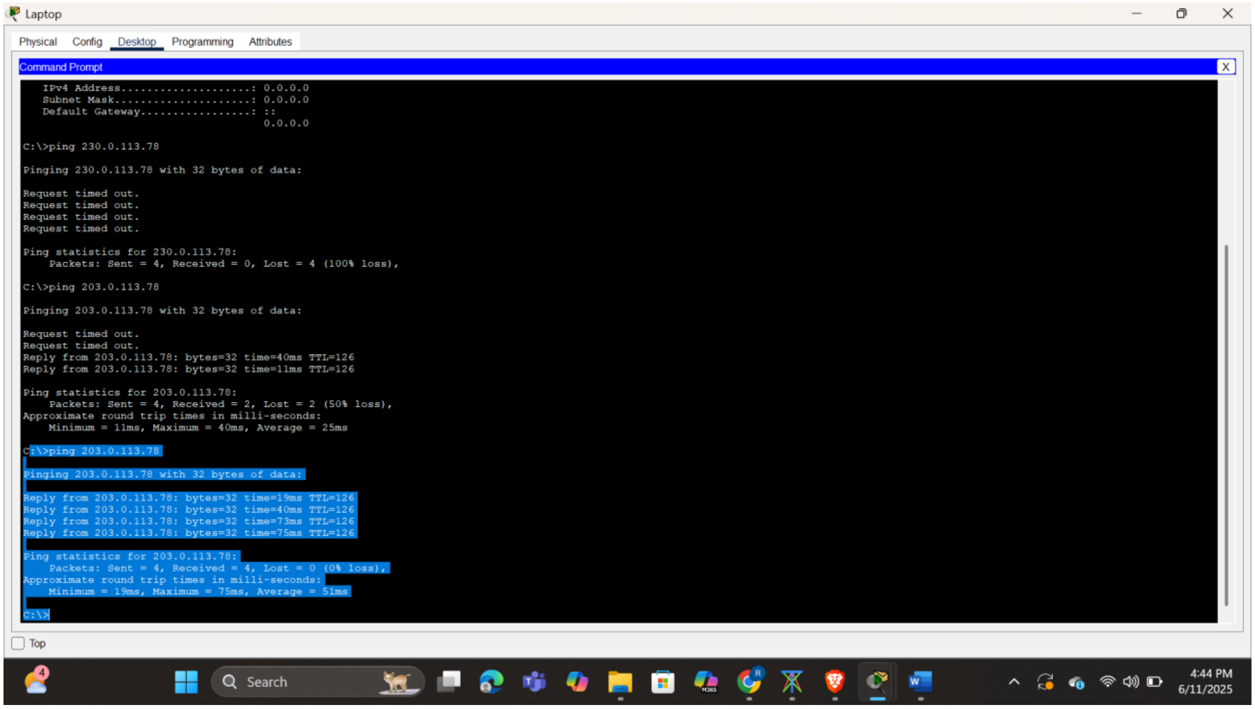

Fig 2.0 Pinging webserver from pc

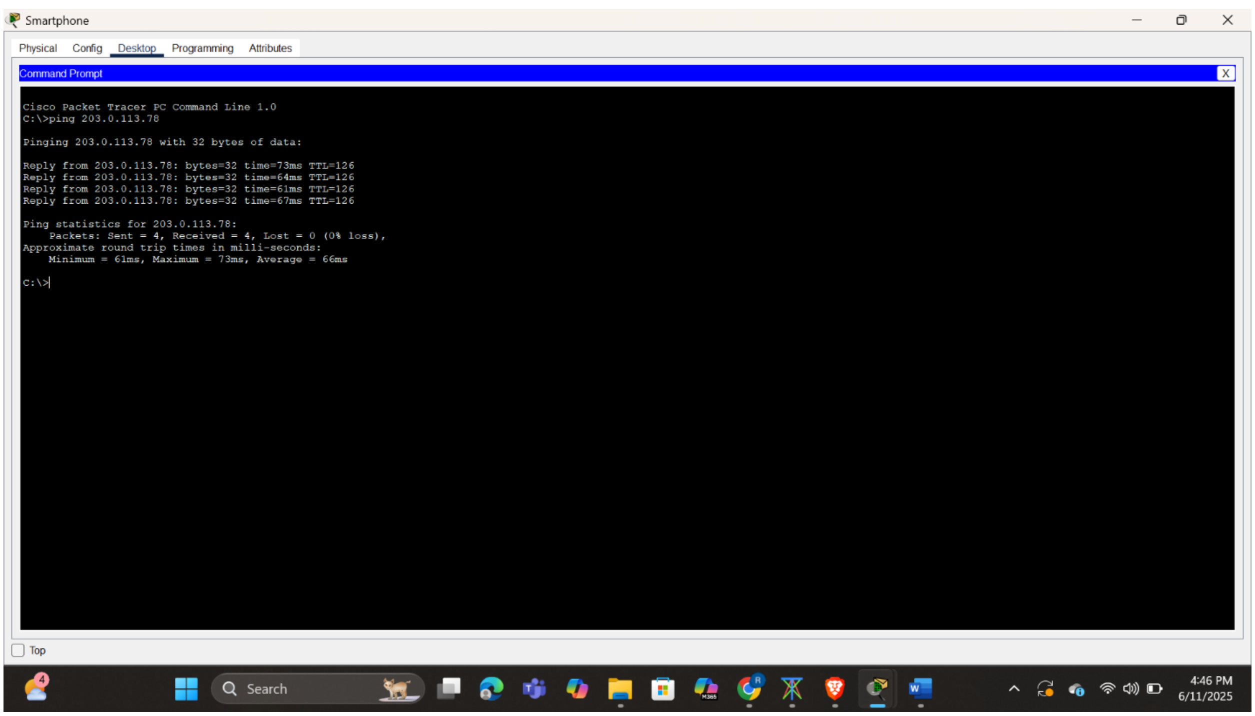

Fig 2.1 Pinging webserver from smartphone

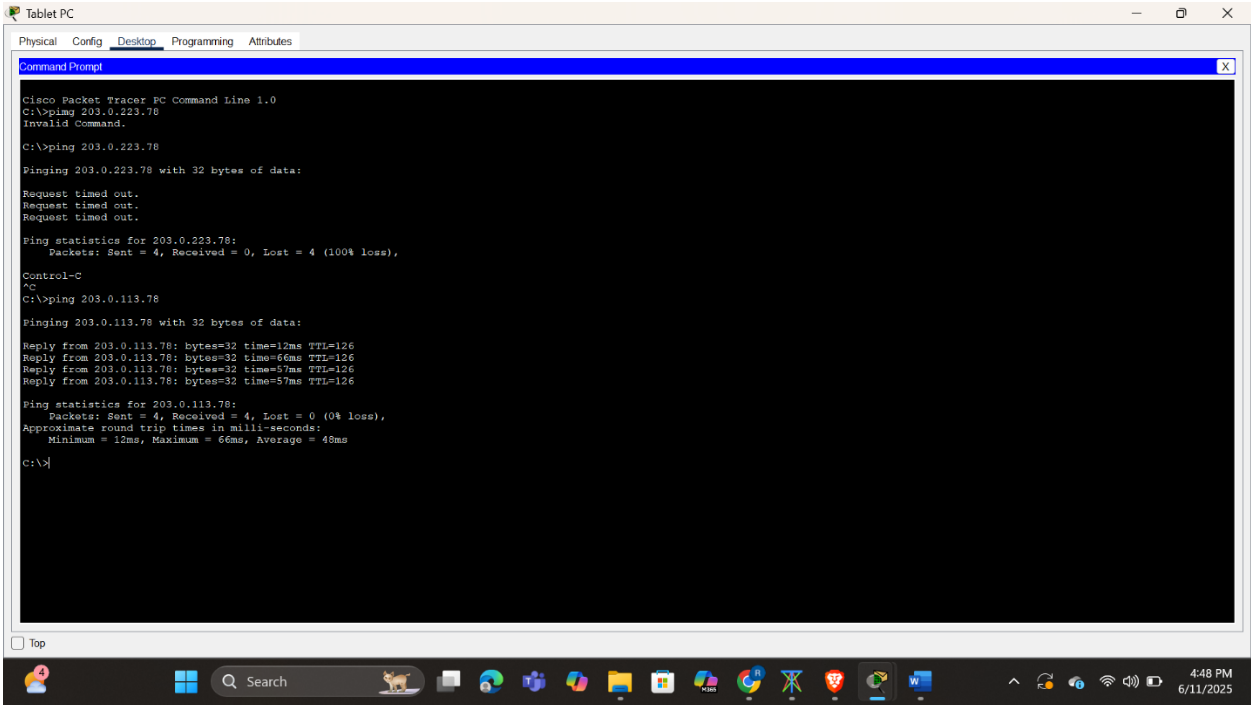

Fig 2.2 Pinging webserver from tablet pc

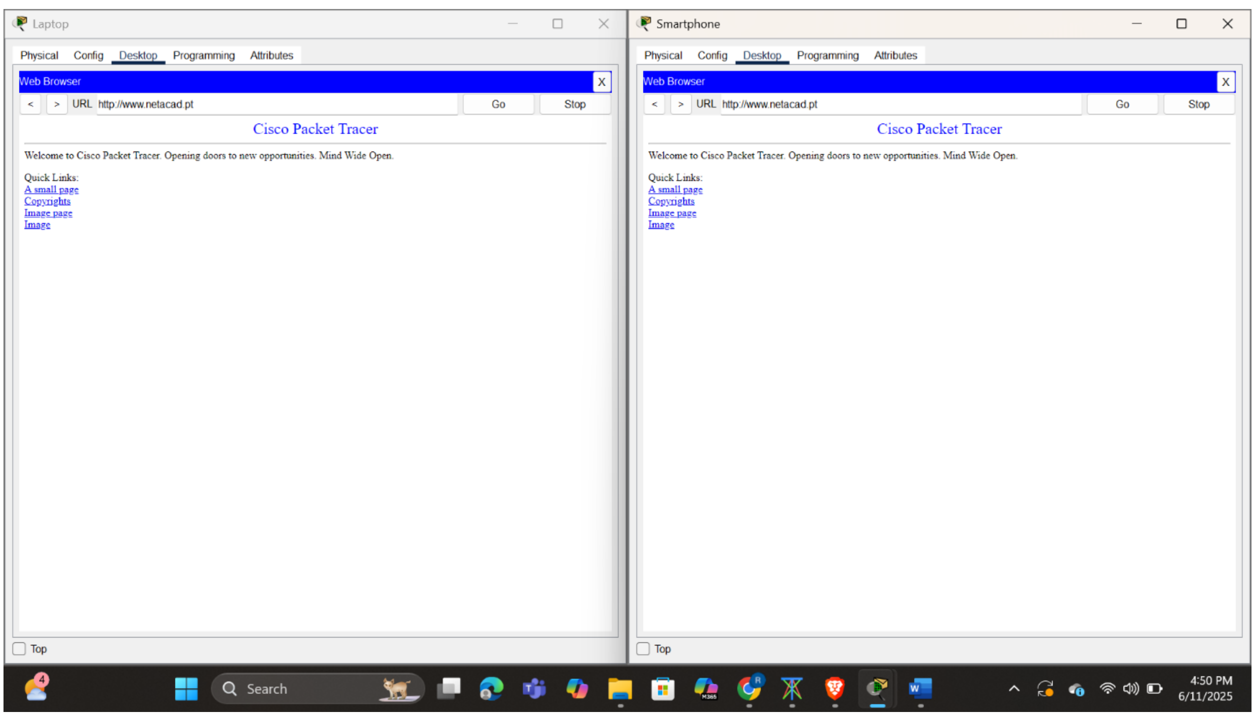

Fig 2.3 Visiting web server from browsers in pc and smartphone

Configuring WLC Controller Network

Configuring VLAN interfaces.

To begin configuring the enterprise wireless infrastructure, I accessed the WLC-1 management interface via a web browser from the Enterprise Admin PC. After successfully logging in with the provided default credentials (admin / Cisco123), I proceeded to configure two essential VLAN interfaces—each mapped to a different wireless network segment.

First, I created an interface named WLAN 2. I assigned it VLAN ID 2 and bound it to Port Number 1. I then set the interface IP address to 192.168.2.254 with a subnet mask of 255.255.255.0. As required, I used 192.168.2.1, which corresponds to RTR-1's G0/0/0.2 interface, as both the default gateway and the primary DHCP server for this network.

Next, I configured the WLAN 5 interface with VLAN ID 5, also on Port 1. This interface was given the IP address 192.168.5.254, with the same subnet mask as before. For its gateway and DHCP server, I specified 192.168.5.1, the IP address of RTR-1's G0/0/0.5 interface.

This step ensured that both VLANs have proper Layer 3 interfaces on the controller and are set up to forward DHCP requests via their respective default gateways. These interfaces are foundational for enabling wireless access and connectivity in the enterprise environment.

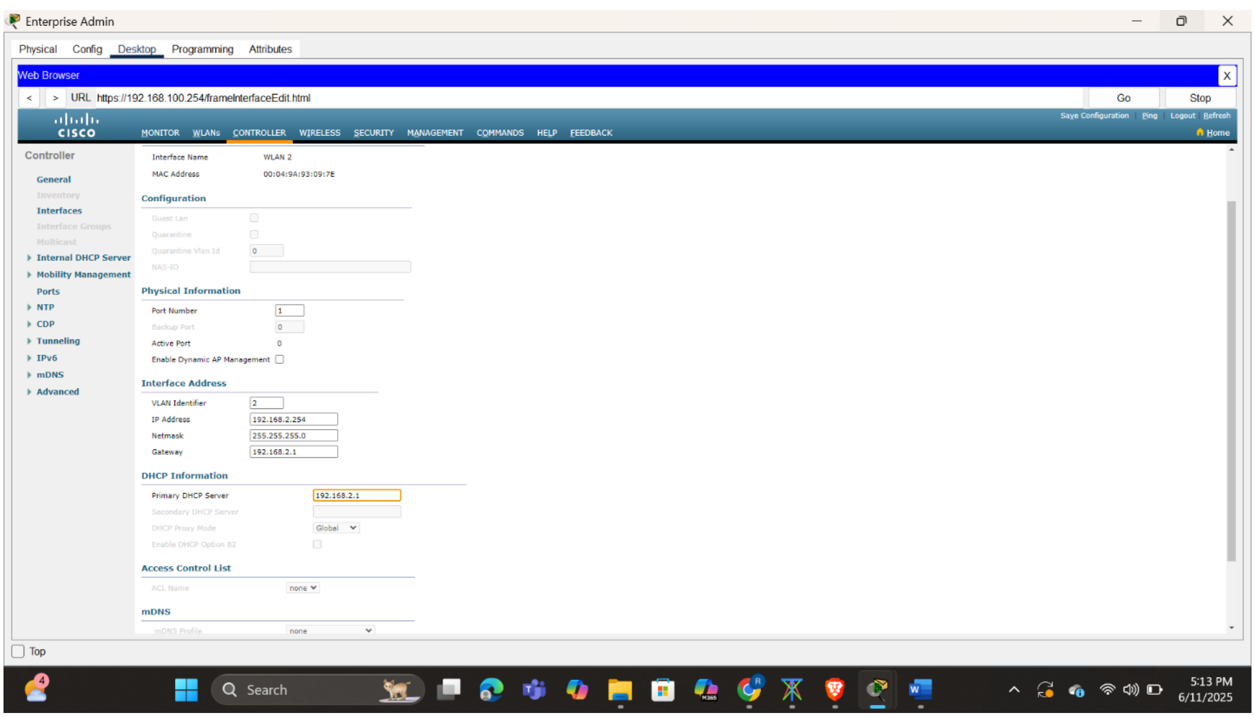

Fig 2.4 Configuring WLAN 2 interface

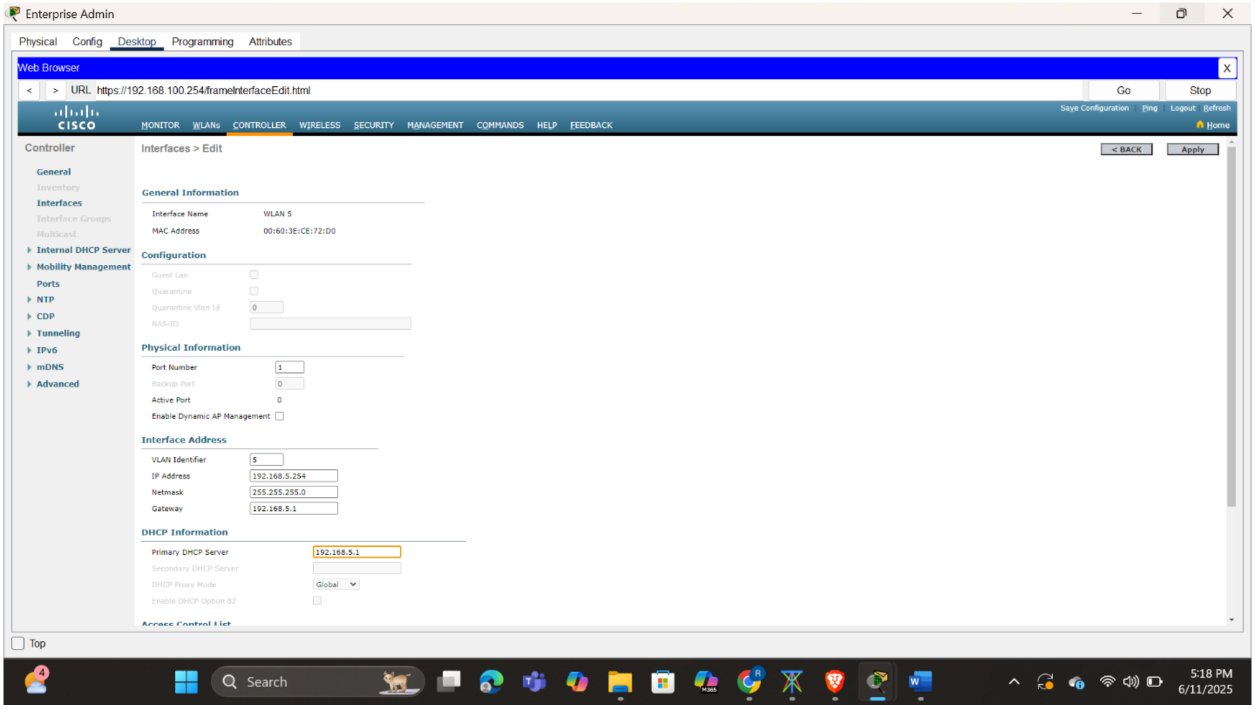

Fig 2.5 Configuring WLAN 5 interface

Configuring DHCP Scope for Wireless Management Network

After successfully configuring the VLAN interfaces on the Wireless LAN Controller (WLC), I moved on to set up a DHCP scope to enable dynamic IP address allocation specifically for the wireless management network, which operates on VLAN 100; this was an essential step to ensure that wireless infrastructure devices such as Lightweight Access Points (LAPs) and administrative clients could automatically receive IP addresses without manual intervention. Using the WLC web interface, I created a new DHCP scope named “management” and carefully entered the required parameters: I set the pool start address to 192.168.100.235 and the end address to 192.168.100.245, defined the network as 192.168.100.0 with a netmask of 255.255.255.0, and assigned the default gateway as 192.168.100.1, which corresponds to the IP address of RTR-1’s G0/0/0.100 subinterface. Once all details were correctly filled in, I made sure to enable the DHCP scope so it could start serving IP addresses immediately. This setup empowered the WLC to function as the DHCP server for the management VLAN, allowing connected LAPs and admin clients to join the network seamlessly and communicate efficiently with the controller. As a result, the 192.168.100.0/24 subnet gained a robust and reliable mechanism for automatic IP allocation, supporting centralized management and enhancing the scalability and operational integrity of the WLAN environment.

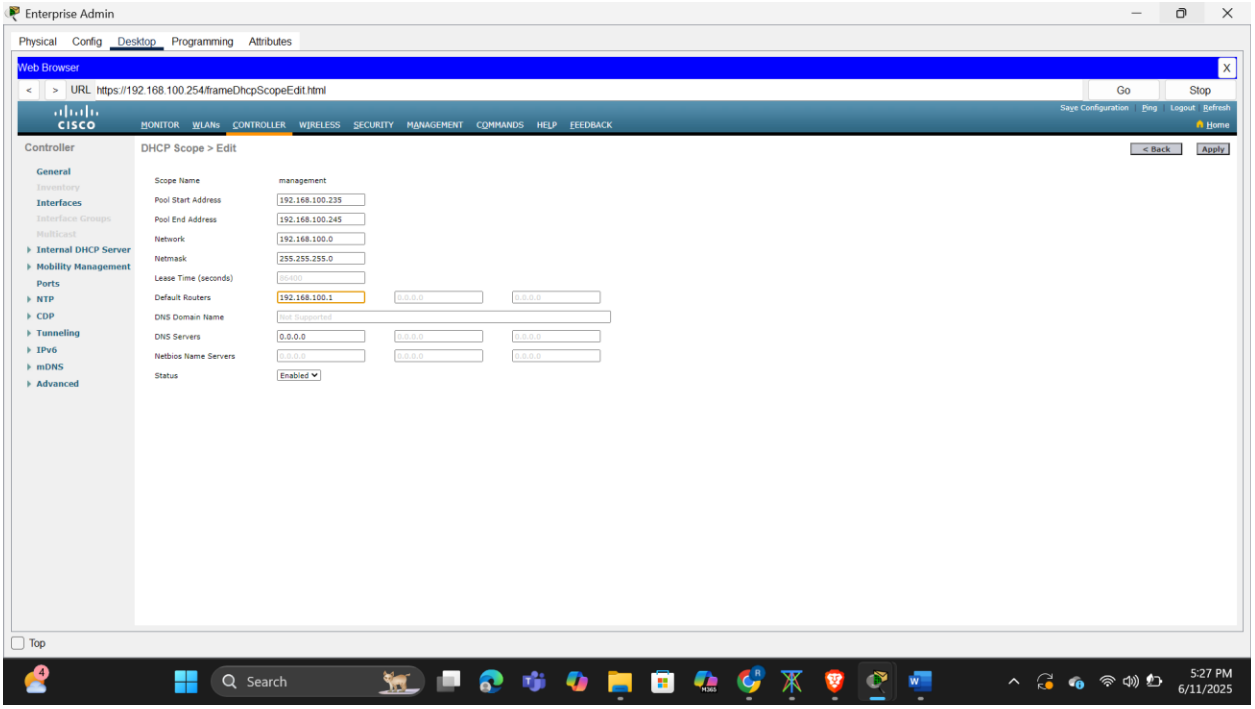

Fig 2.6 Configuring dhcp scope for management

Configuring the WLC with external server addresses.

To establish a secure and centralized authentication and monitoring system within the enterprise WLAN setup, I configured the Wireless LAN Controller (WLC) to integrate with external servers, beginning with the setup of a RADIUS server for 802.1X-based user authentication. Navigating to the WLC's **Security** tab, I accessed the AAA (Authentication, Authorization, and Accounting) settings and created a new RADIUS authentication server entry by assigning it a **Server Index** of 1, specifying the **Server IP address** as 10.6.0.254, and entering the **Shared Secret** as "RadiusPW". This configuration enabled the WLC to authenticate users connecting to the enterprise WLAN (SSID-5) via the external RADIUS server, thus enforcing user identity verification with centralized credentials—an essential feature for enterprise-level WPA2-Enterprise security.

Next, I configured the **SNMP (Simple Network Management Protocol)** server settings to allow the WLC to send system logs and real-time performance data to a centralized monitoring system. Within the same **Security** section, under SNMP settings, I defined the **Community Name** as "WLAN" and provided the **SNMP Server IP address** as 10.6.0.254, which is the same server also hosting the RADIUS service. By completing this configuration, the WLC could now forward system alerts, interface statistics, and event logs to the SNMP server—enabling IT administrators to monitor the network's health and quickly respond to anomalies or performance issues. Together, these steps ensured the WLAN infrastructure not only supported secure user access through authenticated login but also maintained visibility and control through centralized network monitoring and alerting, which are vital for scalable and secure enterprise wireless deployments.

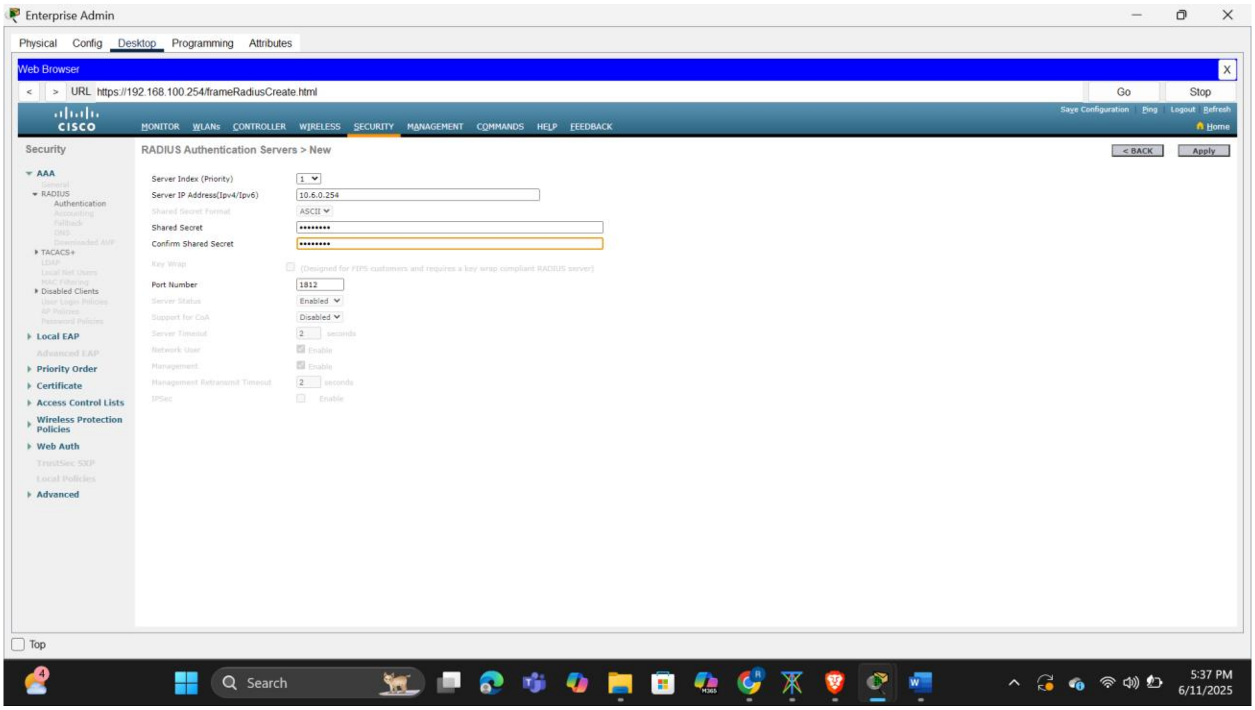

Fig 2.7 Configuring WLC with external server addresses

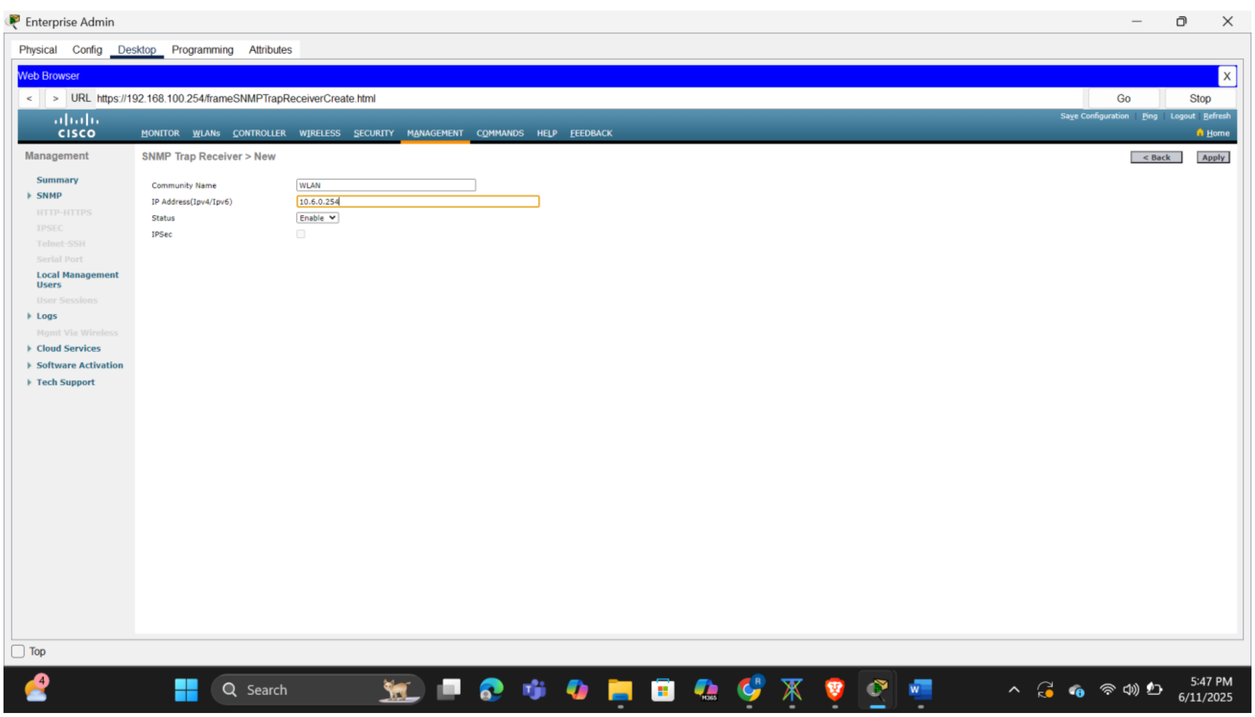

Fig 2.8 Configuring the WLC to send logs information to an SNMP server

Configuring VLAN Interfaces

To begin configuring the WLANs on the Wireless LAN Controller (WLC), I navigated to the **WLANs** tab and clicked **"New"** to initiate the creation of the first wireless network. I assigned **"Wireless VLAN 2"** as the Profile Name, entered **"SSID-2"** as the SSID, and specified the **WLAN ID** as 2. After creating the WLAN, I selected **WLAN 2** from the **Interface Group** dropdown—this interface had been previously configured and associated with VLAN 2. Before proceeding further, I enabled the WLAN by checking the **Status** checkbox and clicking **Apply**, which ensured the network would become operational once all settings were saved.

With the basic setup complete, I proceeded to configure the security settings. Under the **Security** tab, I selected **WPA + WPA2** as the Layer 2 security method. Within the **WPA Policy** section, I enabled **Pre-Shared Key (PSK)** authentication and set the passphrase to **Cisco123**, as specified in the WLAN configuration table. This approach ensured that only devices with the correct passphrase could access the network, providing a secure WPA2-PSK environment suitable for small office or departmental use.

To further enhance performance and localize traffic handling, I moved to the **Advanced** tab, located the **FlexConnect** section, and enabled both **FlexConnect Local Switching** and **FlexConnect Local Authentication**. Enabling these features allowed the lightweight access points (LAPs) to switch traffic locally—reducing dependency on the controller for every packet—and perform authentication locally when the connection to the controller is temporarily unavailable. This setup provided both security and efficiency, especially beneficial for distributed branch deployments where bandwidth and controller availability may be limited.

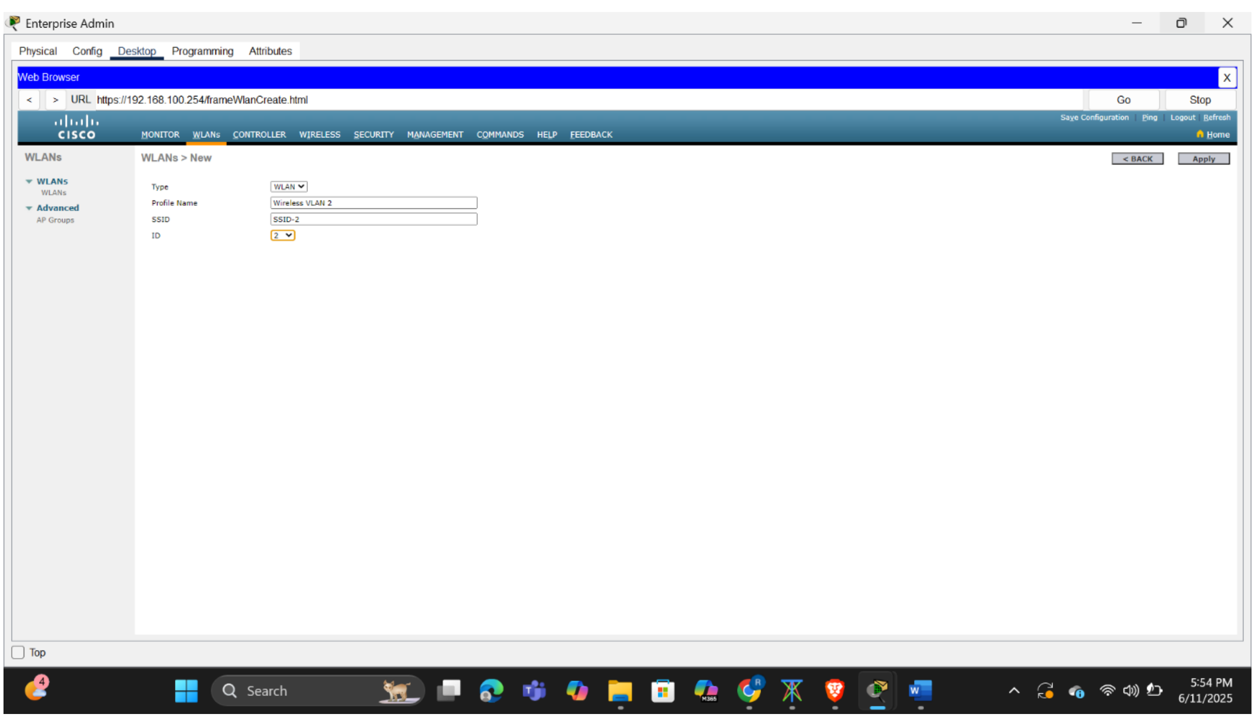

Fig 2.9 Creating first wlan

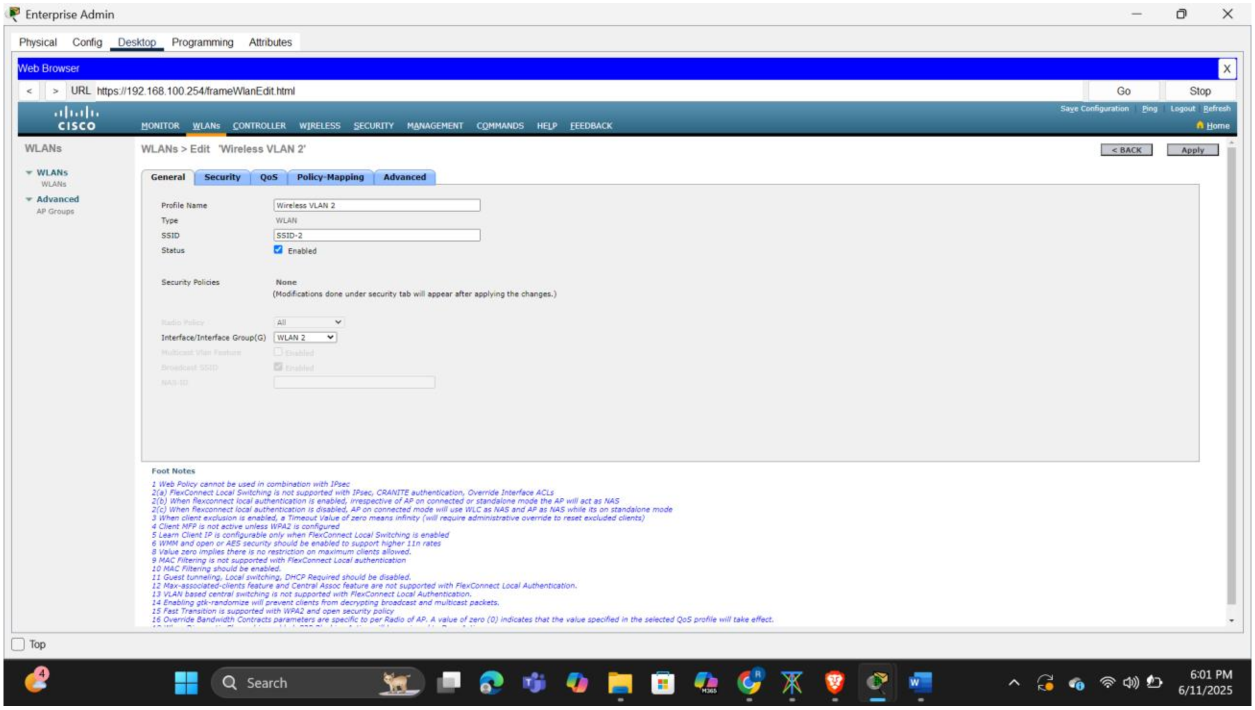

Fig 3.0 Selecting wlan interface

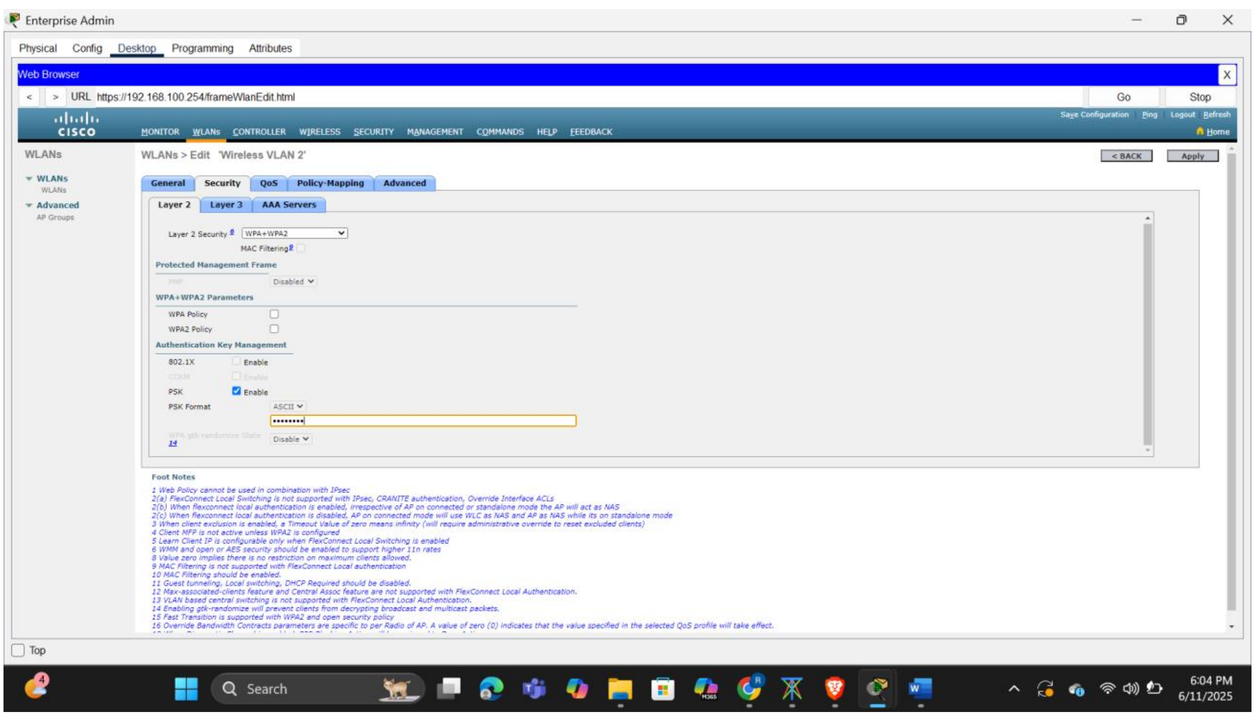

Fig 3.1 Setting up security

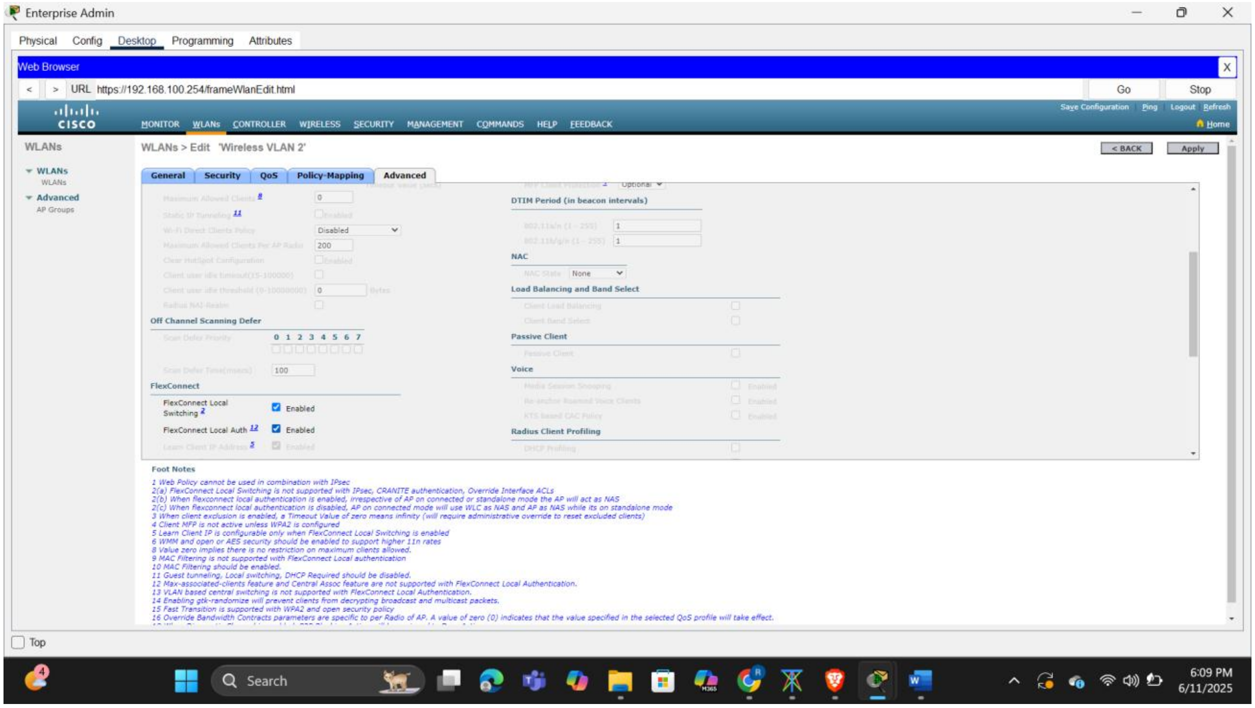

Fig 3.2 Enabling flex connect

Configuring the second WLAN

To configure the second WLAN intended for enterprise use, I returned to the **WLANs** tab on the Wireless LAN Controller and clicked **"New"** to define the new wireless network. I set the **Profile Name** to **"Wireless VLAN 5"**, the **SSID** to **"SSID-5"**, and the **WLAN ID** to **5**. From the **Interface Group** dropdown, I selected **WLAN 5**, which was previously configured and mapped to VLAN 5. I then enabled the WLAN by checking the **Status** box and clicked **Apply** to save the configuration, ensuring the network would be available for connection.

Next, I moved to the **Security** tab to implement enterprise-level security. Under **Layer 2 Security**, I chose **WPA + WPA2**, and within the **WPA Policy** section, I selected **802.1X** authentication instead of PSK. This option requires client devices to authenticate using credentials via a RADIUS server, providing a much higher level of security appropriate for enterprise environments.

To link the WLAN to the **RADIUS server**, I selected the RADIUS authentication server I had previously configured in Step 3 (with IP address 10.6.0.254 and the shared secret "RadiusPW"). This ensured that any user attempting to connect to **SSID-5** would have their credentials verified by the centralized RADIUS server, enforcing user-specific access control and logging.

To complete the configuration and optimize network performance and resilience, I went to the **Advanced** tab and accessed the **FlexConnect** section. I enabled both **FlexConnect Local Switching** and **FlexConnect Local Authentication**, allowing traffic to be switched locally at the access point and enabling continued authentication functionality even if the WLC connection becomes temporarily unavailable. This setup provided a secure, scalable, and resilient enterprise WLAN that leverages centralized authentication while maintaining local efficiency.

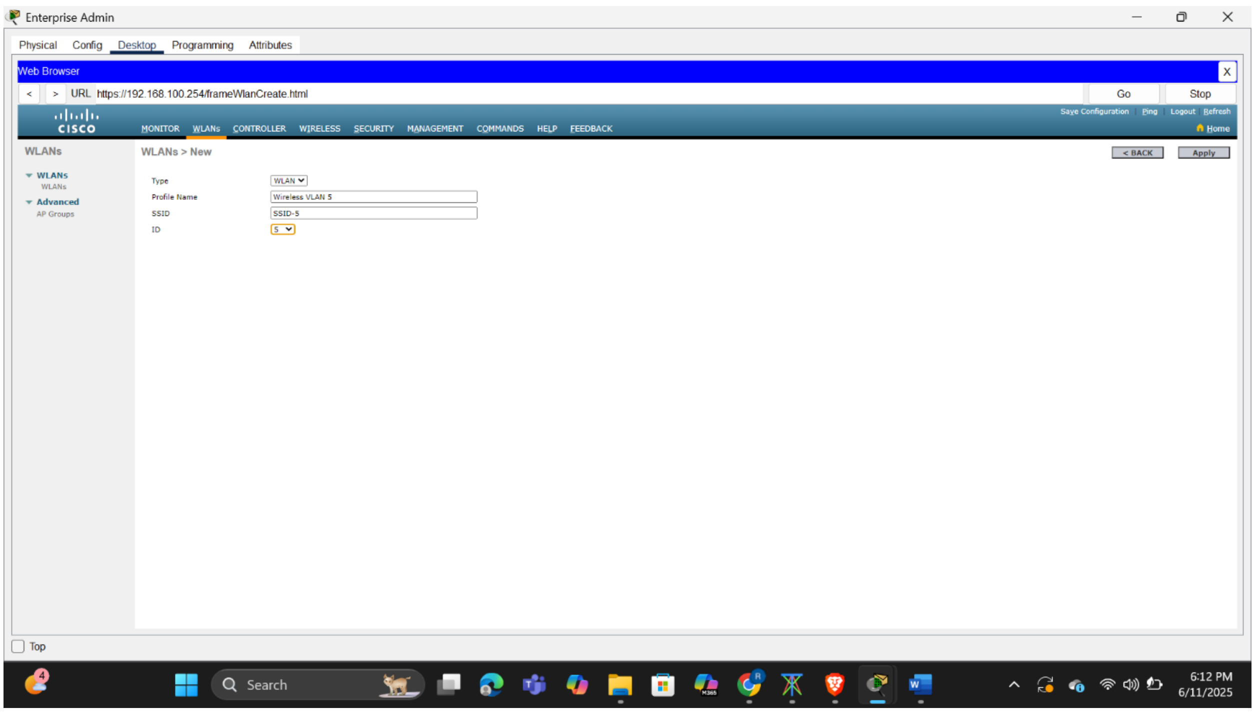

Fig 3.3 Creating wlan 5

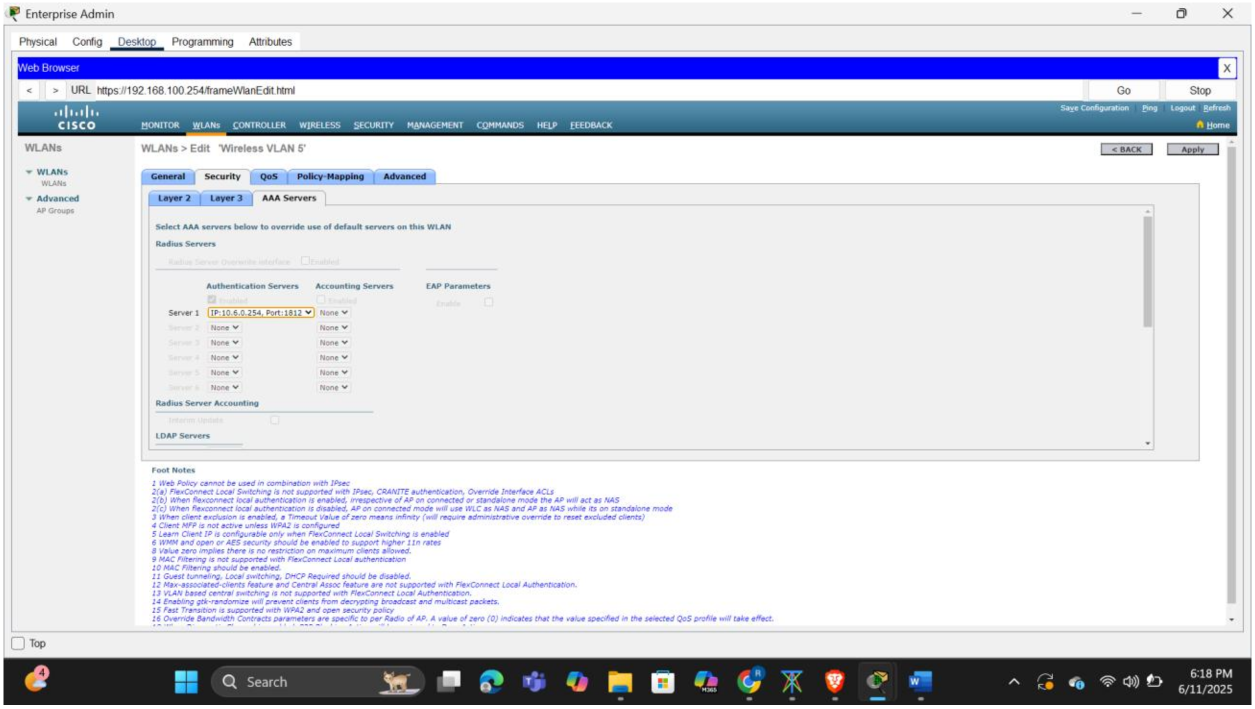

Fig 3.4 Configuring wlan 5 to use radius server authentication

Configuring the Hosts to Connect to the WLANs

To begin connecting the wireless hosts to the respective WLANs, I attempted to detect the broadcast SSIDs using the PC Wireless app on both Wireless Host 1 and Wireless Host 2. However, I noticed that the SSIDs were not initially visible during the wireless scan. After some troubleshooting, I discovered that the issue stemmed from a certificate or SSID broadcast limitation due to date mismatch—likely affecting time-based authentication or visibility policies.

To resolve this, I changed the system date on the WLC and the client devices to January 20th, 2025. Once the date was updated and synchronized across devices, I was immediately able to detect SSID-2 and SSID-5 in the available networks list.

With Wireless Host 1, I connected to SSID-2 by launching the PC Wireless app, selecting the SSID, and entering the pre-shared key (Cisco123) as configured during WLAN creation. The connection was successful, and the host obtained an IP address via DHCP.

For Wireless Host 2, I also opened the PC Wireless app, connected to SSID-5, and when prompted, entered the enterprise credentials—username: userWLAN5 and password: userW5pass. Authentication was handled by the RADIUS server configured earlier. Upon successful verification, the host received a valid IP address and joined the network.

This step confirmed that both WPA2-PSK and WPA2-Enterprise configurations were correctly implemented and functioning as intended, with SSIDs now visible and accessible due to the corrected system date.

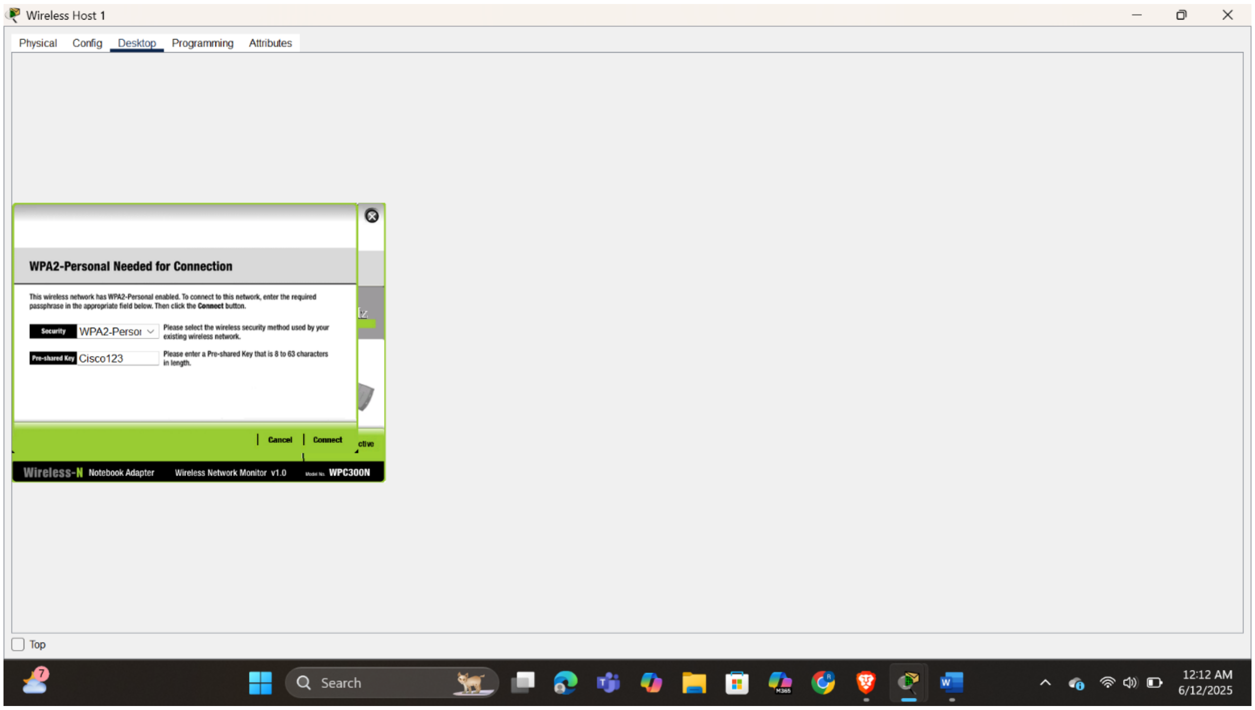

Fig 3.5 Connecting wireless host 1 to WLAN 2

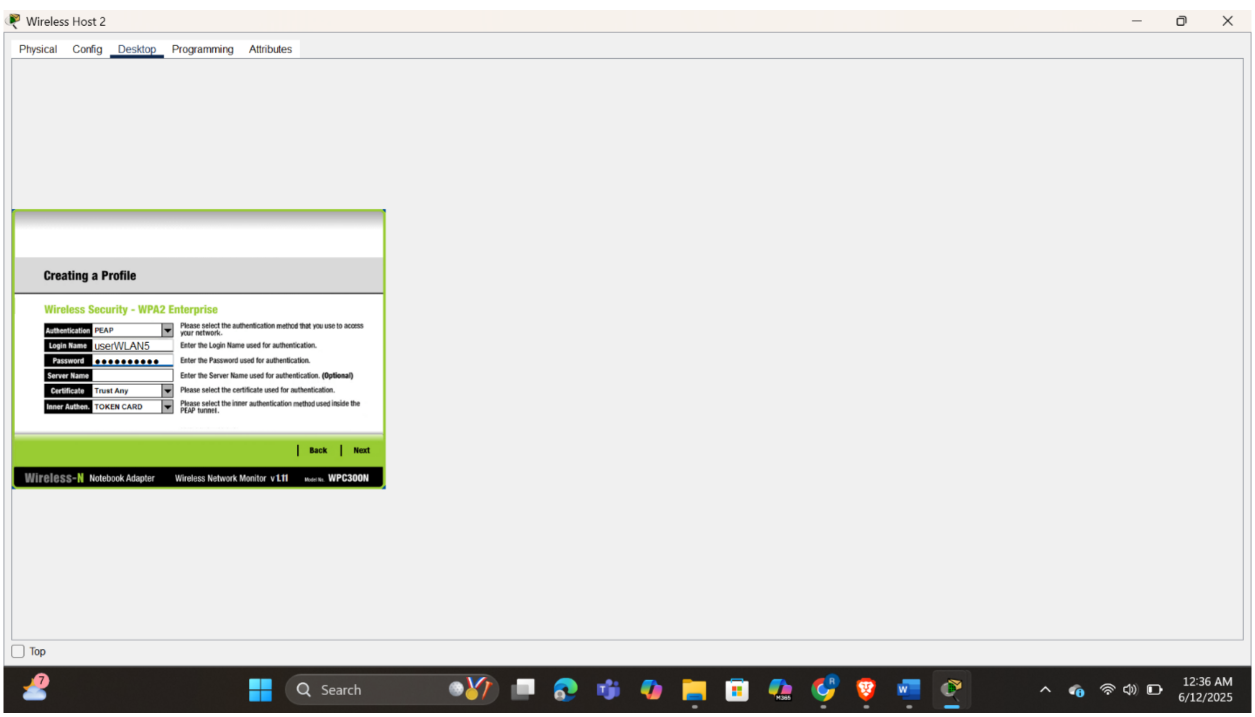

Fig 3.6 Connecting wireless host 2 to wlan 5

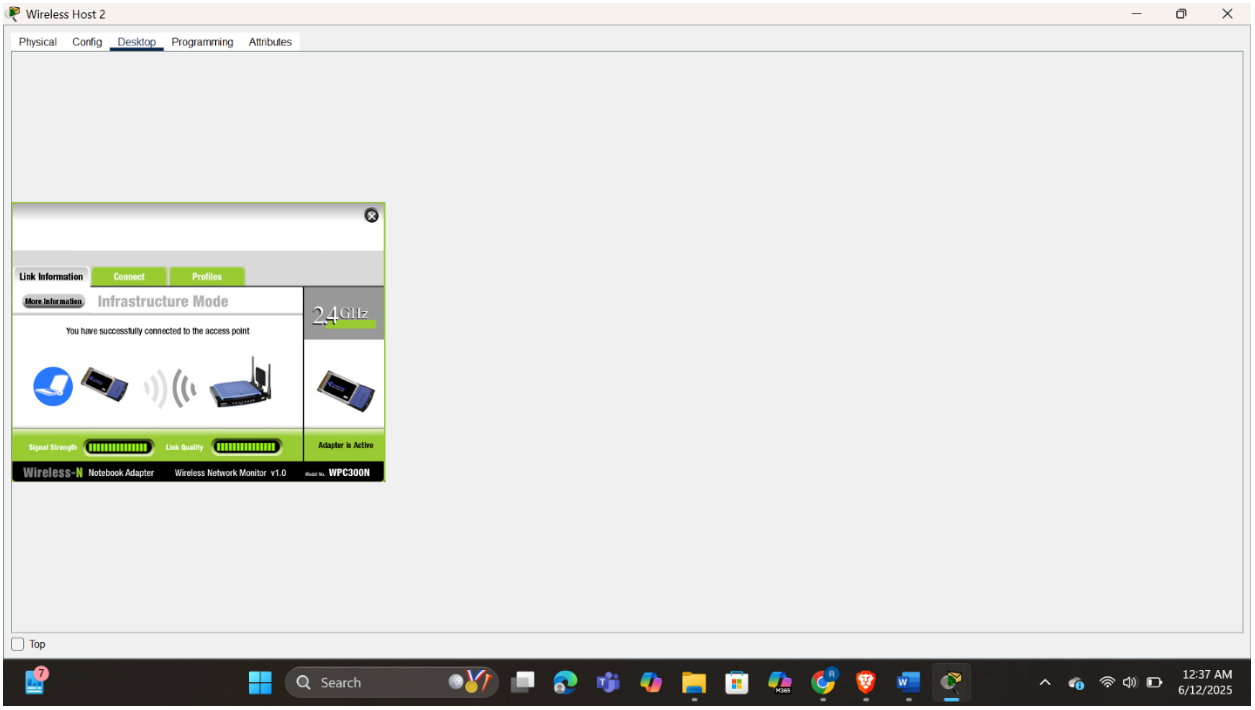

Fig 3.7 Successful connection to wlan 5

Testing connectivity

To finalize the WLAN configuration and verify network functionality, I initiated connectivity tests by pinging the web server from both wireless hosts. From Wireless Host 1, which was connected to SSID-2 (WPA2-PSK), I successfully sent ICMP echo requests to the web server's IP address (203.0.113.78) and received timely replies, confirming layer 3 connectivity. Similarly, Wireless Host 2, authenticated via SSID-5 (WPA2-Enterprise), was also able to reach the web server using both ping and by accessing the web server's URL via a browser, which confirmed successful DNS resolution and HTTP connectivity. These tests validated that both WLANs were functioning correctly, DHCP and RADIUS services were operational, and end-to-end connectivity was achieved as designed.

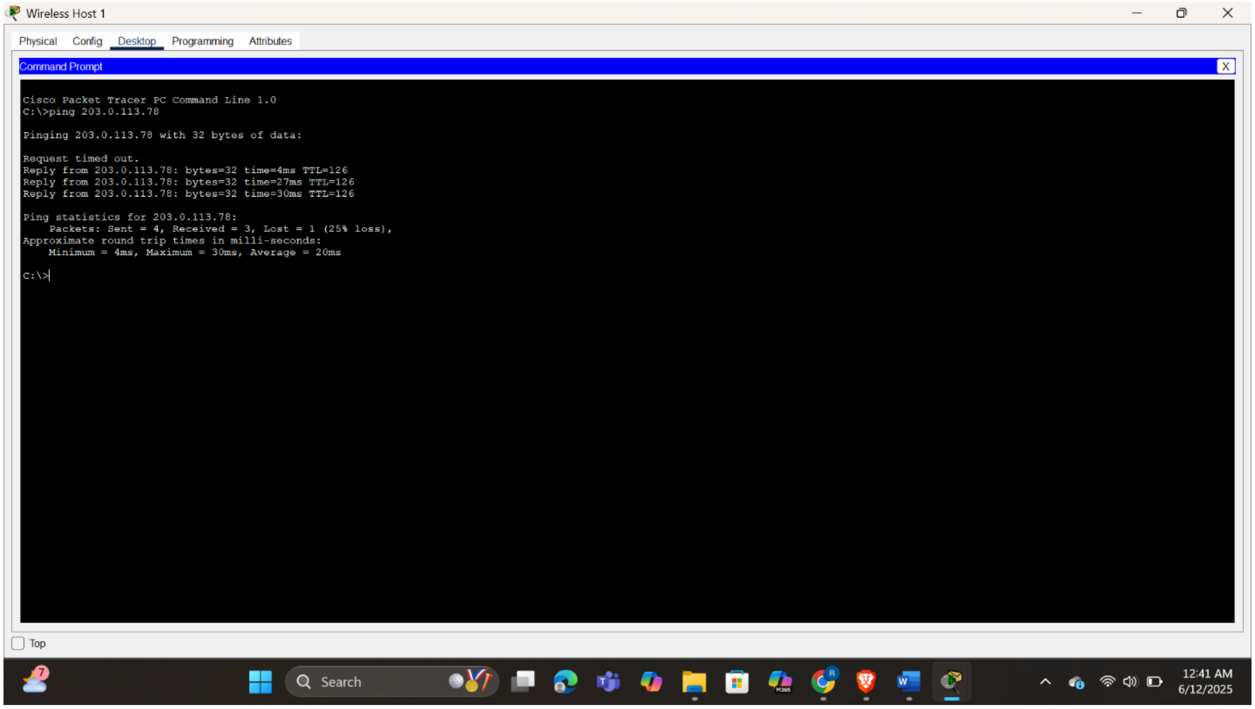

Fig 3.8 Pinging webserver from host 1

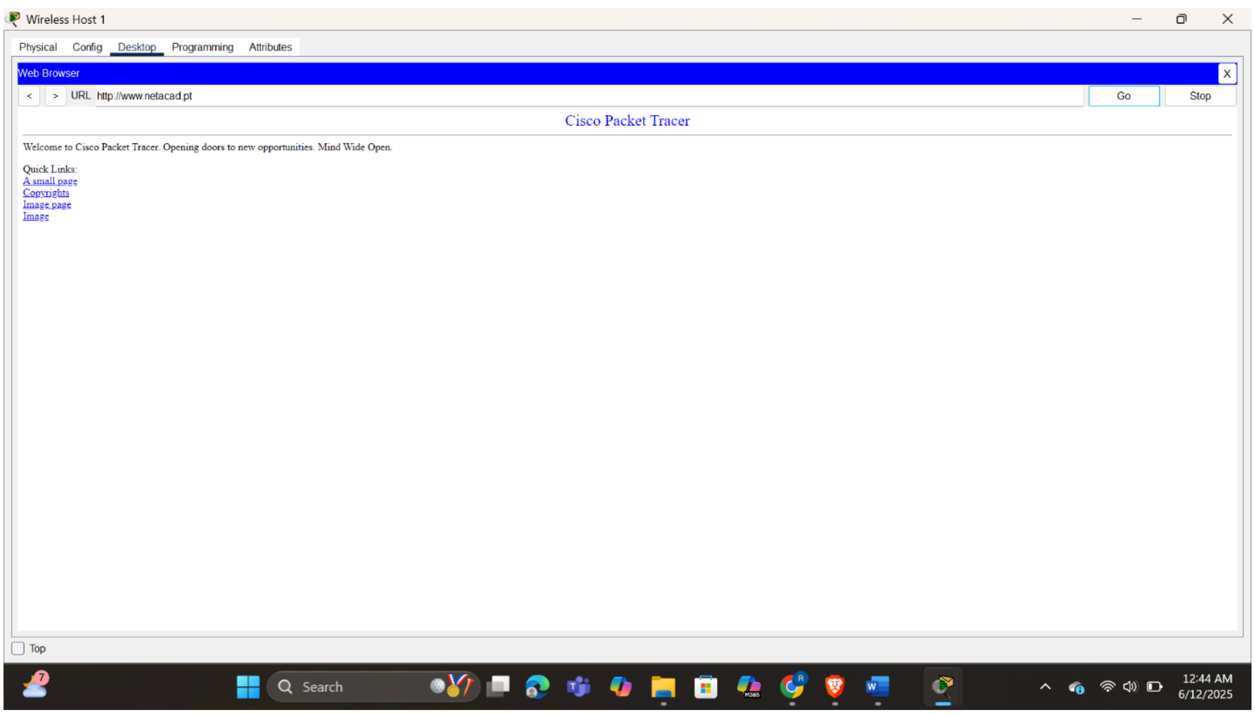

Fig 3.9 Accessing webserver from host 1 web server

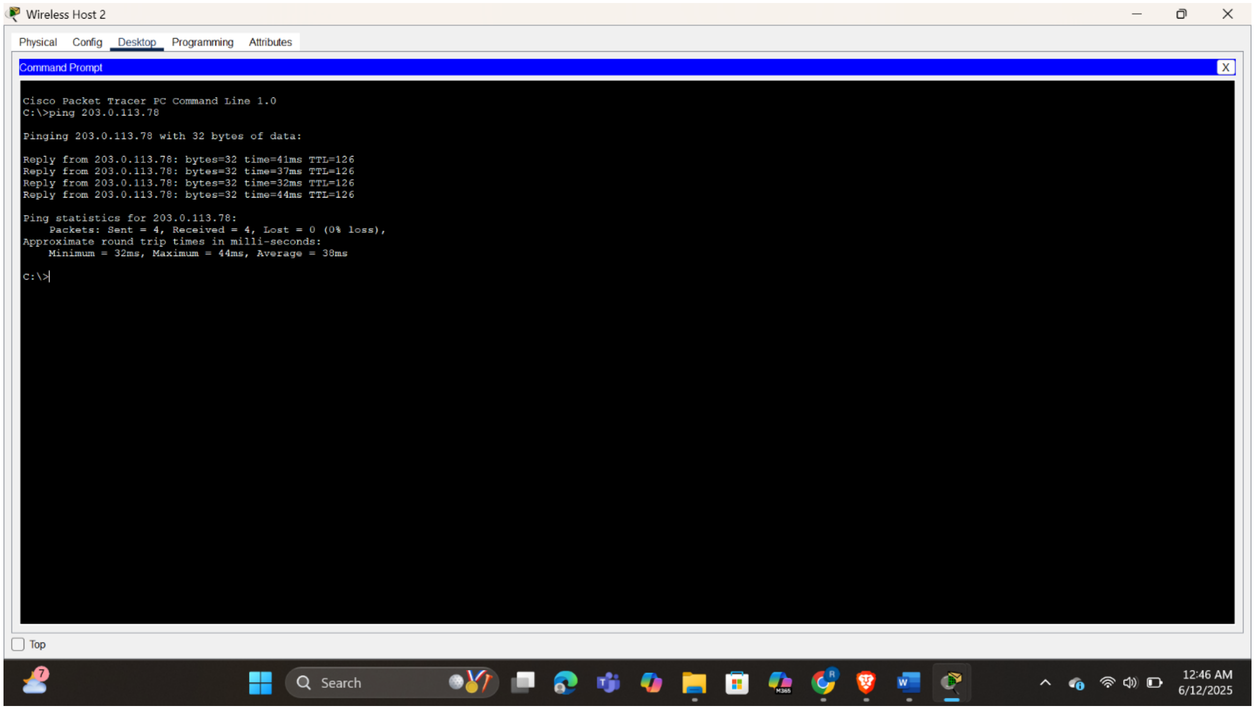

Fig 4.0 Pinging web server from wireless host 2

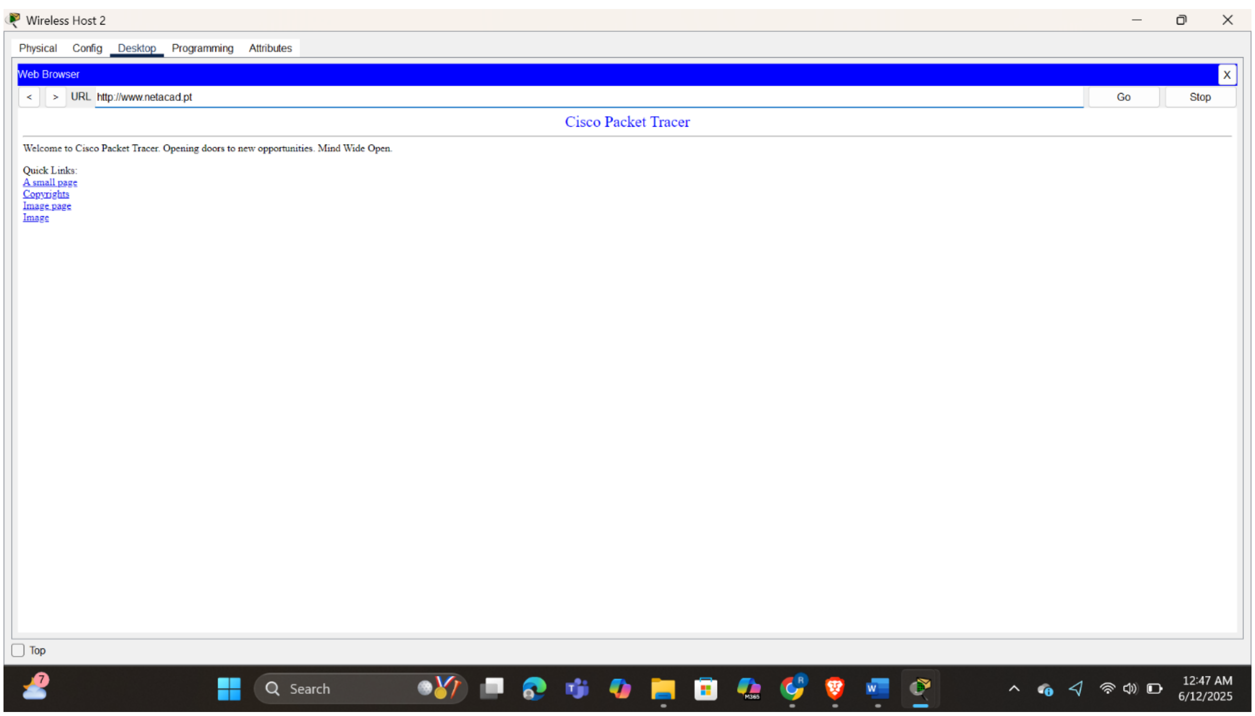

Fig 4.1 Accessing web server from wireless host 2 web browser

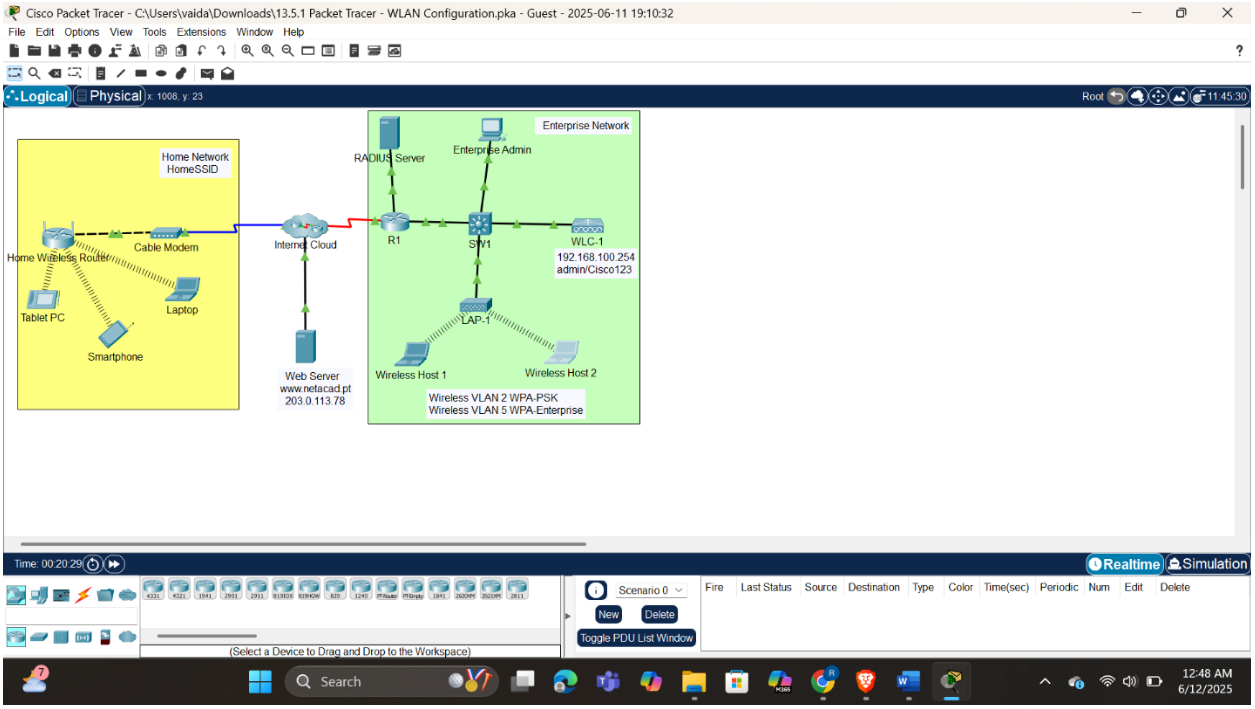

Fig 4.2 Final look with all devices connected

Conclusion

Through this lab, I gained practical experience in configuring wireless networks with varying security levels and management structures. I successfully set up a home wireless router with DHCP and WPA2-Personal, then progressed to enterprise-level configurations using a WLC to manage multiple VLANs and WLANs. I integrated RADIUS server authentication for enterprise-grade security and ensured proper client connectivity and IP assignment. Troubleshooting challenges—like SSID visibility tied to system date—enhanced my problem-solving skills and deepened my understanding of time-sensitive components in WLAN environments. Overall, this lab effectively bridged theoretical knowledge with practical implementation, preparing me to design and manage secure wireless networks in both home and corporate settings.